Solutions

Algebraic problems

Question 9.1:

(b) Write out the auxiliary equations:

$$8\lambda^2 - 2\lambda - 1 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times (-1) \times 8}}{2 \times 8} = \frac{2 \pm \sqrt{36}}{16} = \frac{2 \pm 6}{16}$$

so the roots are $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = -\frac{1}{4}$. So as the roots are real and distinct the general solution is:

$$y(t) = Ae^{\frac{t}{2}} + Be^{-\frac{t}{4}}$$

(c) Write out the auxiliary equations:

$$4\lambda^2 - 24\lambda + 27 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-24) \pm \sqrt{(-24)^2 - 4 \times 4 \times 27}}{2 \times 4} = \frac{24 \pm \sqrt{144}}{8} = \frac{24 \pm 12}{8}$$

so the roots are $\lambda_1 = \frac{9}{2}$ and $\lambda_2 = \frac{3}{2}$. So as the roots are real and distinct the general solution is:

$$y(t) = Ae^{\frac{9t}{2}} + Be^{-\frac{3t}{4}}$$

(d) Write out the auxiliary equations:

$$\lambda^2 - 4 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{0 \pm \sqrt{0^2 - 4 \times 1 \times (-4)}}{2 \times 1} = \frac{0 \pm \sqrt{16}}{2}$$

so the roots are $\lambda_1 = -2$ and $\lambda_2 = 2$. So as the roots are real and distinct the general solution is:

$$y(t) = Ae^{2t} + Be^{-2x}$$

Question 9.2:

(a) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{2t} + Be^{-8x}$$

We know that $y = 2$ and $\frac{dy}{dx} = -6$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{2t} + Be^{-8x} \\ \frac{dy}{dx} &= 2Ae^{2t} - 8Be^{-8x} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= 2 \\ 2A - 8B &= -6 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} 2(A+B) - (A-8B) = 4+6 \implies 6B = 6 \quad &\implies \quad B = 1 \\ A + B = 2 \implies A + 1 = 2 \quad &\implies \quad A = 1 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = e^{2t} + e^{-8x}$$

(b) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{\frac{t}{2}} + Be^{-\frac{t}{4}}$$

We know that $y = 10$ and $\frac{dy}{dx} = 2$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{\frac{t}{2}} + Be^{-\frac{t}{4}} \\ \frac{dy}{dx} &= \frac{A}{2}e^{\frac{t}{2}} - \frac{B}{4}e^{-\frac{t}{4}} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= 10 \\ \frac{A}{2} - \frac{B}{4} &= 2 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} A + B + 4\left(\frac{A}{2} - \frac{B}{4}\right) = 10 + 8 \implies 3A = 18 \quad &\implies \quad A = 6 \\ A + B = 10 \implies 6 + B = 10 \quad &\implies \quad B = 4 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = 6e^{\frac{t}{2}} + 4e^{-\frac{t}{4}}$$

(c) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{\frac{9t}{2}} + Be^{-\frac{3t}{4}}$$

We know that $y = 0$ and $\frac{dy}{dx} = 12$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{\frac{9t}{2}} + Be^{-\frac{3t}{4}} \\ \frac{dy}{dx} &= \frac{9A}{2}e^{\frac{9t}{2}} - \frac{3B}{4}e^{-\frac{3t}{4}} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= 0 \\ \frac{9A}{2} - \frac{3B}{2} &= 12 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} 3(A+B) + 2(\frac{9A}{2} - \frac{3B}{2}) = 0 - 24 \implies 12A = 24 \quad &\implies \quad A = 2 \\ A + B = 0 \implies 2 + B = 0 \quad &\implies \quad B = -2 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = 2e^{\frac{9t}{2}} - 2e^{-\frac{3t}{4}}$$

(d) From Question 9.1(a), the general solution is:

$$y(t) = Ae^{2t} + Be^{-2x}$$

We know that $y = -4$ and $\frac{dy}{dx} = 0$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{2t} + Be^{-2x} \\ \frac{dy}{dx} &= Ae^{2t} - 2Be^{-2x} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= -4 \\ 2A - 2B &= 0 \end{aligned}$$

This leaves a system of simultaneous equations which we can solve:

$$\begin{aligned} 2(A+B) + 2A - 2B = -8 + 0 \implies 4A = -8 \quad &\implies \quad A = -2 \\ A + B = 0 \implies -2 + B = -4 \quad &\implies \quad B = -2 \end{aligned}$$

Then the full general solution is given by:

$$y(t) = -2\left(e^{2t} + e^{-2x}\right)$$

Question 9.3: Write out the auxiliary equations:

$$3\lambda^2 + 2\lambda - 8 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4 \times 3 \times (-8)}}{2 \times 3} = \frac{-2 \pm \sqrt{100}}{6} = \frac{-2 \pm 10}{6}$$

so the roots are $\lambda_1 = \frac{4}{3}$ and $\lambda_2 = -2$. So as the roots are real and equal the general solution is:

$$y(x) = Ae^{\frac{4t}{3}} + Be^{-2t}$$

We know that $y = -6$ and $\frac{dy}{dx} = -18$ when $x = 0$. Write out y and $\frac{dy}{dx}$:

$$\begin{aligned} y &= Ae^{\frac{4t}{3}} + Be^{-2t} \\ \frac{dy}{dx} &= \frac{4A}{3}e^{\frac{4t}{3}} - 2Be^{-2t} \end{aligned}$$

Plug in the initial conditions into y and $\frac{dy}{dx}$:

$$\begin{aligned} A + B &= -6 \\ \frac{4A}{3} - 2B &= -18 \end{aligned}$$

Substitute to solve:

$$\begin{aligned} 2(A+B) + \frac{4A}{3} - 2B = -12 - 18 \implies \frac{10A}{3} = -30 \quad &\implies \quad A = -9 \\ A + B = -6 \implies -9 + B = -9 \quad &\implies \quad B = 3 \end{aligned}$$

Then the full general solution is given by:

$$y = 3e^{-2t} - 9Ae^{\frac{4t}{3}}$$

Question 9.4: Write out the auxiliary equations:

$$\lambda^2 - 4\lambda + 4 = 0$$

Find the roots of the equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times 4}}{2 \times 1} = \frac{4 \pm \sqrt{0}}{2}$$

so the roots are $\lambda_1 = \lambda_2 = 2$. So as the roots are real and equal the general solution is:

$$y(x) = (A + Bx)e^{2x}$$

Real-world problems

Question 9.5: Newtons laws of motion (as 2nd Order ODEs):

(a) The total forces on the object are $F = mg\sin(30^\circ)$. We integrate to get the velocity (v) of this object (v from 0 to v and t from 0 to t):

$$\cancel{m}\frac{dv}{dt} = \cancel{m}g\sin(30^\circ) \implies \int_0^v dv = \frac{g}{2}\int_0^t dt \implies [v]_0^v = \frac{g}{2}[t]_0^t \implies v(t) = \frac{g}{2}t$$

We integrate again to get the distance travelled (x) of this object (x from 0 to x and t from 0 to t):

$$v = \frac{dx}{dt} = \frac{g}{2}t \int_0^x dx = \int_0^t \frac{g}{2}tdt \implies [x]_0^x = \frac{g}{2}\left[\frac{t^2}{2}\right]_0^t \implies x(t) = \frac{g}{4}t^2$$

So the velocity increases linearly and the distance quadratically in this problem.

(b) The total forces on the object are $F = mg\sin(30^\circ) - mg\cos(30^\circ)$. We integrate to get the velocity (v) of this object (v from 0 to v and t from 0 to t):

$$\begin{aligned}
&\cancel{m}\frac{dv}{dt} = \cancel{m}g\sin 30^\circ - \cancel{m}g\cos 30^\circ \\
\implies &\int_0^v dv = g(\sin(30^\circ) - \cos(30^\circ))\int_0^t dt \\
\implies &[v]_0^v = g(\sin(30^\circ) - \cos(30^\circ))[t]_0^t \\
\implies &v(t) = g(\sin(30^\circ) - \cos(30^\circ))t
\end{aligned}$$

We integrate again to get the distance travelled (x) of this object (x from 0 to x and t from 0 to t):

$$\begin{aligned}
&v = \frac{dx}{dt} = \frac{g}{2}g(\sin(30^\circ) - \cos(30^\circ))t \\
\implies &\int_0^x dx = g(\sin(30^\circ) - \cos(30^\circ))\int_0^t tdt \\
\implies &[x]_0^x = \frac{g}{2}(\sin(30^\circ) - \cos(30^\circ))\left[\frac{t^2}{2}\right]_0^t \\
\implies &x(t) = \frac{g}{2}(\sin(30^\circ) - \cos(30^\circ))t^2
\end{aligned}$$

Again, the velocity increases linearly and the distance quadratically in this problem.

Question 9.6: Fick's Law of diffusion (as 2nd Order ODEs): We need to solve:

$$\frac{d^2C}{dx^2} = 0$$

Differentiating once gives us:

$$\frac{dC}{dx} = A$$

Differentiating twice gives us:

$$C(x) = Ax + B$$

In general, the boundary conditions that $C(0) = C_0$ and $C(1000) = C_{1km}$ means that

$$\begin{aligned} C(0) &= B = C_0 \\ C(1000) &= 1000A + C_0 = C_{1km} \implies A = \frac{C_{1km} - C_0}{1000} \end{aligned}$$

Therefore the solution to Fick's law is:

$$C(x) = C_0 + \frac{C_{1\,km} - C_0}{1000} x$$

Using $C_0 = 1$ and $C_{1km} = 0.1$, houses 500m away have concentration

$$\begin{aligned} C(x) &= C_0 + \frac{C_{1\,km} - C_0}{1000} x \\ &= 1 + \frac{0.1 - 1}{1000} \times 500 \\ &= 0.55 \end{aligned}$$

So the concentration is 0.55mg/m^{-3}, which is above the 0.5 limit.

Question 9.7: Fouriers law of heat transfer (as 2nd Order ODEs): This is similar to Question 9.6. We need to solve:

$$\frac{d^2T}{dx^2} = 0$$

Differentiating once gives us:

$$\frac{dT}{dx} = A$$

Differentiating twice gives us:

$$T(x) = Ax + B$$

In general, the boundary conditions that $T(0) = T_0$ and $T(3000) = T_{3km}$ means that

$$\begin{aligned} T(0) &= B = T_0 \\ T(3000) &= 3000A + T_0 = T_{3km} \implies A = \frac{T_{3km} - T_0}{3000} \end{aligned}$$

Therefore the solution to Fick's law is:

$$T(x) = T_0 + \frac{T_{3km} - T_0}{3000} x$$

Plugging in $T_0 = 10$ and $T_{3km} = 250$ gives

$$T(x) = 10 + \frac{250 - 10}{3000} x = 10 + \frac{2}{25} x$$

Question 9.8: Newtons laws of motion (as 2nd Order ODEs): Integrate once

$$\begin{aligned}
\frac{d}{dt}\left(\frac{dx}{dt}\right) &= a \\
\int_0^{\frac{dx}{dt}} d\left(\frac{dx}{dt}\right) &= \int_0^t a dt \\
\frac{dx}{dt} &= at + B
\end{aligned}$$

When $t = 0$ we know that $v = v_0$ so then we see that $B = v_0$ then

$$\frac{dx}{dt} = at + v_0$$

Integrate a second time:

$$\begin{aligned}
\frac{dx}{dt} &= at + v_0 \\
\int_0^x dx &= \int_0^t at + v_0 dt \\
x(t) &= \frac{a}{2}t^2 + v_0 t + C
\end{aligned}$$

When $t = 0$ we know that $x = 0$ so then we see that $C = 0$ then

$$x(t) = \frac{a}{2}t^2 + v_0 t$$